

CSS 385 A: Introduction to Game Development

Spring 2021 | Monday/Wednesday 3:30PM-5:30PM

Instructor: Dr. Yusuf Pisan, pisan@uw.edu UW1-260Q (425) 352-3741

Office Hours: Monday 1-2pm and Thursday 2-3pm. Signup via [Canvas Calendar](#)

Class Discord: [invite link](#)

Course Description

Examines the fundamental issues in designing and developing computer video games; creative and artistic elements, story narration, software architecture, interaction model, mathematic, physics, special effects, and in-game AI logic. Experiences elements in game design: world setting, game play, and interface; and experiences implementing games: conceptualization, prototyping, and play testing.

Learning Objectives

- Critically examine video games
- Understand the structure of games
- Design, prototype, test and implement a game from scratch
- Understand and extend techniques commonly used in games
- Work in groups, present and reflect on extended project

Textbooks

- None - Readings will be provided

Grading

- Exercises: 20%
- Projects: 40%
- Final Progress: 10%
- Final Project: 30%

A scale of 90s (3.5■4.0), 80s (2.5■3.4), 70s (1.5■2.4), 60s (0.5■1.4) is a rough guide. A student who achieves 75% of the possible points will receive a 2.0 grade in this course. Students with 95% and above will receive a 4.0 grade. Scores in between will

be interpolated.

Distance Learning

Owing to the UW response to COVID-19, this course will be delivered entirely online through the Canvas learning management system and Zoom (supplemented by other web resources). This means that you will need to put in more time on your own than in a conventional course. Of course, the benefit of this structure is that you do not need to be physically present at UW-Bothell.

This course is scheduled to run synchronously at your scheduled class time via Zoom. These Zoom class sessions will be recorded. The recording will capture the presenter's audio, video and computer screen. Student audio and video will be recorded if they share their computer audio and video during the recorded session. The recordings will only be accessible to students enrolled in the course to review materials. These recordings will not be shared with or accessible to the public. The University and Zoom have FERPA-compliant agreements in place to protect the security and privacy of UW Zoom accounts. Students who do not wish to be recorded should 1) change their Zoom screen name to hide any personal identifying information such as their name or UW netid, and 2) not share their computer audio or video during their Zoom sessions.

Policies

Pet Policy: If a pet enters the camera frame during class, we will pause our discussion for an introduction to that pet and admiration by all. All pets are welcome.

Assignment Submission: You can miss one exercise without penalty.

For projects, you can work individually or in pairs. When working as a pair

- Both students must enter a comment in the Canvas assignment with their name and their partner's name 72 hours before the assignment is due. Submit an empty file if necessary to leave a comment.
- All substantial work must be performed with both students present (virtually).

You can submit one and only one project 24 hours late without any penalty (if working as a pair both people will need to use their one-time extension). If you are using your 24 hour extension, you must post a brief explanation to "Using Extension" assignment before the assignment due date. I will make allowances for exceptional circumstances such as sickness, bereavement and official university business. I will not make exceptions for work, other classes, personal obligations, etc.

If you have any questions about your assignment grade, you must contact me within one week.

Academic Integrity: Other than when working in pairs, talking about code is OK, looking at each others code is not OK. Looking at references to understand how a functions gets used is OK; looking up assignment solutions is not OK. It is also not acceptable for your code or solutions to be publicly accessible on the web (for example, a public GitHub repository). Plagiarism will result in an assignment score of zero and a misconduct letter in your student record. Please be very careful to adhere to the student code of conduct:

<http://www.washington.edu/cssc/for-students/student-code-of-conduct/> Any code or text that you copy from an external source must have a comment (or reference) indicating its origins.

Attendance: Attend all classes. You are responsible for all the material covered in class, as well as any announcements including change of due dates or assignment specifications. There will also be graded in-class group exercises to practice problem solving. If you miss a class, I expect you to make-up for it on your own by asking your friends, reviewing the textbook, lecture materials, etc.

Communication: We will use discord as an extension of the classroom. Use a meaningful nickname and act professionally. If your question can be answered publicly by me or by a classmate, post it to discord. Use the office hours for complex issues or topics you are struggling with.

Use your UW email rather than "Canvas Messaging" to communicate directly with me. "Canvas Submission Comments" should only be used to draw the grader's attention to a specific part of your submission.

Problems: If you are having difficulties, come and talk to me. If I don't know about it, I cannot help you. Small problems can be fixed easily early in the quarter, but might become impossible to fix later on.

Course Material: Lecture notes and other material will be posted to Canvas under "Files".

See the [School of STEM Course Policies](#), which covers:

- Academic Integrity

- Access and Accommodations
- Classroom Emergency Preparedness
- For Our Veterans
- Grade of Incomplete
- Inclement Weather
- Parenting Resources
- Religious Accommodations
- Respect for Diversity
- Student Support Services
- Surviving Sexual and Relationship Violence
- Wonder How to Address Faculty?

The UW Religious Accommodations Policy can be found at
<https://registrar.washington.edu/staffandfaculty/religious-accommodations-policy/>

Course Calendar

Topics and due dates below might be modified as needed.

See [Weekly Notes](#) for additional readings and pointers to external resources

Week	Topic	Notes
Week-1	Introduction, Game Mechanics	Unity Basics
Week-2	Player Experience, Systems	Fun with the tools and the Hero
Week-3	Gameplay, Prototyping	Next Generation Hero
Week-4	Conceptual Prototype	
Week-5	Game AI	
Week-6	Digital Prototype-1	
Week-7	Digital Prototype-2	
Week-8	Alpha Playtest	
Week-9	Beta Playtest	
Week-10	Final Demos, Wednesday June 2	Monday, 31 May, Memorial Day